Probabilistic point-source cross-matching: state of the art, gaps, and a proposal for FLINT

A hardcore-level technical research report — compiled April 2026

Executive summary (BLUF)

The probabilistic cross-matching of astronomical point sources is undergoing a regime change. Budavári–Szalay (2008) Bayes factors plus the Pineau et al. (2017) χ-match formalism have powered every operational pipeline of the last decade — CDS X-Match, NWAY, Marrese's Gaia external match, ARCHES — but they assume Gaussian astrometry, factorisable photometric priors, and treat *association*, *classification*, *deblending*, and *novelty* as separate downstream problems. The Rubin/LSST alert stream (~10^7 alerts/night, first scientific alerts 24 Feb 2026), Gaia DR4 (~December 2026, 2.5 G sources with full epoch series), arXiv Euclid DR1 (Oct 2026), and Roman (NET Sep 2026) will overwhelm this stack. The frontier has moved to amortised neural posterior estimation (sbi 2.x, ltu-ili, SimBIG), normalising-flow population priors, and contrastive multi-survey foundation models (AstroCLIP, Multimodal Universe). **No existing framework unifies association + classification + novelty detection + selection-aware demographics in a single, calibrated, streaming-capable Bayesian computation.** The third part of this report proposes one — provisionally named **FLINT** (Flow-Learned Identification and Novelty in Time-domain cross-matching) — and the fourth gives Victor a concrete, dated 18-month roadmap to land a first-author paper in *MNRAS* or *Astronomy and Computing* by 2027 via collaboration with CBPF, IAG-USP, or the Fink-Brazil broker team.

PART 1 — STATE OF THE ART

1. The Budavári–Szalay framework and its N-catalogue extensions

The modern field starts with the **likelihood-ratio estimator** of Sutherland & Saunders (1992, *MNRAS* 259, 413). :antCitation[]{citations="10a1bdde-ab7d-49e7-9a6b-e3806c59693b"} For a primary detection and a candidate counterpart with magnitude m at uncertainty-normalised separation r:

$$LR(m, r) = \frac{q(m)\, f(r)}{n(m)}$$

where $f(r) = \frac{1}{2\pi\sigma_\alpha\sigma_\delta}\exp\big[-\tfrac{1}{2}(\Delta\alpha^2/\sigma_\alpha^2 + \Delta\delta^2/\sigma_\delta^2)\big]$ is the bivariate Gaussian offset p.d.f., $q(m)$ the magnitude distribution of true counterparts, $n(m)$ the surface density of background field sources brighter than m. oup Oxford Academic With prior odds Q that the primary has any identifiable counterpart, the **reliability** of candidate i is

$$R_i = \frac{LR_i}{\sum_j LR_j + (1 - Q)}$$

and operational completeness/reliability curves $C(L_{\rm th})$ and $R(L_{\rm th})$ as functions of an LR threshold trace the canonical trade-off that pervades all later work.

Budavári & Szalay (2008, *ApJ* 679, 301; arXiv:0707.1611) recast cross-matching as symmetric Bayesian model comparison arxiv +2 between hypothesis H (one common source at unknown direction $\boldsymbol{\omega}$) and K (independent sources). Using the Fisher distribution on the sphere IOPscience arXiv $f(\boldsymbol{x}; \boldsymbol{\omega}, \kappa) = \frac{\kappa}{4\pi\sinh\kappa}\exp(\kappa\,\boldsymbol{\omega}\cdot\boldsymbol{x})$ with $\kappa_i = 1/\sigma_i^2$ and uniform isotropic prior on $\boldsymbol{\omega}$, the integrals are analytic, yielding the celebrated **two-catalogue Bayes factor**:

$$\boxed{B_{12} \;=\; \frac{2}{\sigma_1^2 + \sigma_2^2}\,\exp\!\left[-\frac{\psi_{12}^2}{2(\sigma_1^2 + \sigma_2^2)}\right]}$$

The prefactor $2/(\sigma_1^2+\sigma_2^2)$ rewards small joint uncertainty — thresholds on B are *not* equivalent to thresholds on ψ. :antCitation[]{citations="b731bb7c-8e07-4c73-8a3e-2d9dc1cfa42f"} For n catalogues with circular errors,

$$B \;=\; 2^{\,n-1}\,\frac{\prod_i w_i}{\sum_i w_i}\,\exp\!\left[-\,\frac{\sum_{i<j} w_i w_j\,\psi_{ij}^2}{\sum_k w_k}\right], \qquad w_i \equiv 1/\sigma_i^2$$

The posterior follows from prior odds calibrated empirically from local source densities (Heinis, Budavári & Szalay 2009, *ApJ* 705, 739):

$$P(H|D) = \frac{B\,P(H)/P(K)}{1+B\,P(H)/P(K)}, \qquad \frac{P(H)}{P(K)} = \frac{N_*}{(N_1-N_*)(N_2-N_*)}\,\frac{1}{\Delta\Omega}$$

where N_* is iteratively re-estimated (IOPscience) by counting tuples above a tentative posterior cut. Operational thresholds minimise $1-T(P_0)+F(P_0)$ where T is completeness and F the false-acceptance fraction.

Budavári & Loredo (2015, *Annu. Rev. Stat. Appl.* 2, 113; arXiv:1610.04178) generalise this to a **partition framework** :antCitation[]{citations="d35f232b-dd33-4b28-8672-c43b2413261d"} over the union of detections :antCitation[]{citations="3dcbf919-730b-4e10-9876-f6b8ea51105a,5fe3d6ab-f2b8-4904-9add-d7d98c46d593"} $D=\{(i,c)\}$. The marginal likelihood of an object with detection set S_o is $\mathcal{L}_o = \int p(\boldsymbol{\omega})\prod_{(i,c)\in S_o} p(\boldsymbol{x}_{ic}|\boldsymbol{\omega})\,d\boldsymbol{\omega}$:antCitation[]{citations="5d8fe3df-1f96-4f05-90a7-4066d3db4de5"} and the optimal cross-match maximises $\mathcal{L}(\Pi)\,p(\Pi) = \prod_{o\in\Pi}\mathcal{L}_o\cdot p(\Pi)$. **Budavári & Basu (2016, *AJ* 152, 86)** solved the 2-catalogue case with the Hungarian algorithm :antCitation[]{citations="4b2f66c6-effa-4377-bf11-9052c3003189"} in $O(n^3)$; Shi, Budavári & Basu (2019, *ApJ* 870, 51) :antCitation[]{citations="7e2407fe-1683-4d51-90c4-3ccd566762fc"} and Nguyen et al. (2022) extended via integer linear programming :antCitation[]{citations="784a22af-a444-4e10-a27e-147bface3a67,0071adee-9b2a-43dd-b078-d2da3f823b23"} and MIQCP for globally optimal N-catalogue assignments. :antCitation[]{citations="cd99689d-3ec7-46fb-95a1-066b24b22642"}

2. Photometric and SED extensions

Pineau et al. (2017, *A&A* 597, A89; arXiv:1609.00818) — the ARCHES χ-match — promotes positional errors to elliptical 2×2 covariances $\boldsymbol{V}_i$ (aanda) and defines the **generalised Mahalanobis distance**

$$Q_{\chi^2} = \sum_{i=1}^{n}(\boldsymbol{\mu}_i-\boldsymbol{\mu}_\Sigma)^\top \boldsymbol{V}_i^{-1}(\boldsymbol{\mu}_i-\boldsymbol{\mu}_\Sigma), \quad \boldsymbol{V}_\Sigma^{-1} = \sum_i \boldsymbol{V}_i^{-1}$$

with weighted-mean position $\boldsymbol{\mu}_\Sigma = \boldsymbol{V}_\Sigma\sum_i \boldsymbol{V}_i^{-1}\boldsymbol{\mu}_i$. Under the match hypothesis $D_M = \sqrt{Q_{\chi^2}} \sim \chi_{2(n-1)}$:antCitation[]{citations="8b2e13d1-81bc-4490-aea3-4405b9cced18"} and candidates are accepted at confidence $\gamma = 0.9973$ ($k_\gamma = 3.4439$ for $n=2$ — the spherical analogue of the Rayleigh "3σ"). :antCitation[]{citations="4767beae-1b2b-459d-99db-5567fd31dc9d"} The integral that connects this back to BS08 is

$$\int \prod_i \mathcal{N}_{\boldsymbol{\mu}_i,\boldsymbol{V}_i}(\boldsymbol{p})\,d\boldsymbol{p} = \sqrt{\frac{|\boldsymbol{V}_\Sigma|}{\prod_i|\boldsymbol{V}_i|}}\,\frac{\exp(-\tfrac{1}{2}Q_{\chi^2})}{(2\pi)^{n-1}}.$$

Pineau enumerates Bell-number combinatorial hypotheses :antCitation[]{citations="de34430b-61f5-431c-9dc0-c60d0a7169c3,45ae55eb-7476-45ed-8a6c-47f6ccfea03a"} (5, 15, 52, 203, 877 for $n=3..7$) :antCitation[]{citations="357ab6d2-657f-4047-8854-3232ffc6ea34"} for each candidate tuple, :antCitation[]{citations="39ce93f6-a9a2-40a3-8894-cee109b37010"} with frequentist priors on spurious associations $n_F = \rho_1\cdots\rho_k\,\mathcal{V}_{2(k-1)}(k_\gamma)$ derived from local densities.

The **photometric Bayes factor** factorises on independence of position and photometry (Pineau §11):

$$B_{\rm full} = B_{\rm astro} \times B_{\rm phot}, \quad B_{\rm phot} = \frac{\int \pi(\mathcal{T}) \exp[-\tfrac{1}{2}\chi^2_{\rm SED}(\mathcal{T})]\, d\mathcal{T}}{\prod_c n_c(m_c)}$$

with $\chi^2_{\rm SED} = \sum_i (m_i - m_i^{\mathcal{T}})^2/(\sigma^2_{m_i} + \sigma^2_{\mathcal{T},i})$.

Salvato et al. (2018, *MNRAS* 473, 4937; NWAY) operationalise class-conditional priors (NASA HEASARC) (Oxford Academic) for X-ray→IR matching:

$$B^{(c)}_{\rm prior} = \frac{p(\boldsymbol{\theta}_c \mid \text{counterpart})}{p(\boldsymbol{\theta}_c \mid \text{field})}.$$

For X-ray AGN, $p(W1{-}W2 \mid \text{QSO})$ peaks at ~0.8 (Vega) while $p(W1{-}W2 \mid \text{star})$ peaks near 0; this colour leverage typically delivers 95–99% reliability on XMM-COSMOS. :antCitation[]{citations="73d3f72d-4467-486d-8888-f81b5fb1d868"} NWAY explicitly reports a no-counterpart probability :antCitation[]{citations="22ca5000-e129-48b4-bcc6-abace6e33adc"} via $P_c = 1 - \exp(-\rho\,\pi r^2_{\rm PU})$. :antCitation[]{citations="6f8c8f4b-3f9a-485e-9dab-30551fda4af3,f8bbab62-758a-4e67-81d3-eb926915e8a4"} **Wilson & Naylor (2017, *MNRAS* 473, 5570)** developed a *non-parametric* symmetric photometric Bayes factor that uses each catalogue's empirical $\phi_c(m)$ distribution and a matching kernel $f(m_1, m_2|H)$, gaining roughly an order of magnitude in B over astrometry-only. **Medan, Lépine & Hartman (2021, *AJ* 161, 234)** apply the framework to 5.8 M high-PM Gaia DR2 stars × 2MASS/AllWISE/GALEX/RAVE/SDSS/Pan-STARRS, achieving >95% probabilistic matches for 1.7 M K/M dwarfs.

3. Kinematic and temporal extensions

For a Gaia source $(\alpha_0, \delta_0, \varpi, \mu_{\alpha*}, \mu_\delta)$ at t_0=J2016.0, the predicted position at t is

$$\alpha(t) = \alpha_0 + \frac{\mu_{\alpha*}(t - t_0)}{\cos\delta_0}, \quad \delta(t) = \delta_0 + \mu_\delta(t - t_0)$$

with parallax-induced apparent displacement $\Delta\boldsymbol{x}_\varpi = \varpi(\boldsymbol{X}_\oplus \cdot \hat{e}_\alpha, \boldsymbol{X}_\oplus \cdot \hat{e}_\delta)$ and propagated covariance

$$\boldsymbol{V}(t) = \boldsymbol{V}_0 + (t - t_0)^2 \boldsymbol{V}_\mu + 2(t - t_0)\boldsymbol{C}_{0,\mu}.$$

A high-PM star like Barnard's drifts ~160" between 2000 and 2016 — completely missed by 1" cone searches. **Marrese et al. (2017, *A&A* 607, A105; 2019, *A&A* 621, A144; 2022, Gaia DR3 docs ch. 15)** are the canonical Gaia external-match papers, propagating Gaia coordinates to each external catalogue's epoch and scoring candidates by

$$\text{FoM} = \frac{\exp(-\tfrac{1}{2}D^2_M)}{\Sigma_{\rm local}\, \pi r^2_{\rm search}}.$$

Wilson (2023, *MNRAS* 519, 5882) introduces probabilistic PM priors $f(\mu|\ell, b, m)$ for sources lacking measured PMs — central for LSST since Gaia provides PMs for only ~10% of LSST sources. The effective positional variance broadens as $\sigma^2_{\rm eff} = \sigma^2_{\rm pos} + \langle\mu^2\rangle(t - t_0)^2$.

Time-domain matching combines $B_{\rm astro}(t_a)$ at the alert epoch with a *temporal photometric Bayes factor* comparing variability templates against constancy:

$$B_{\rm var} = \frac{\int \pi(\boldsymbol{\theta}_{\mathcal{C}}) \prod_k \mathcal{N}(m_k|\mu_{\mathcal{C}}(t_k;\boldsymbol{\theta}), \sigma_k)\, d\boldsymbol{\theta}}{\prod_k \mathcal{N}(m_k|\bar{m}, \sigma_k)}.$$

Budavári, Szalay & Loredo (2017, *ApJ* 838, 64) formalise multi-epoch fusion: (Internet Archive) directional and flux Bayes factors multiply, enabling sub-threshold probabilistic detection. (Internet Archive) (Mendeley) **Gavras et al. (2023, *A&A* 674, A22)** apply this to Gaia DR3 × ASAS-SN/OGLE/CRTS/ZTF/VSX with a synthetic distance combining positional Mahalanobis and magnitude-residual scores. The **MUWCLASS** pipeline (Yang et al. 2022; arXiv:2403.05068)

cascades Bayesian matching across 4FGL-DR4 → Chandra/XMM → Gaia DR3 + CatWISE2020 + 2MASS + AllWISE for γ-ray counterpart identification. (ResearchGate)

The unifying summary is a **hierarchy of factorisable Bayes factors**:

$$B_{\text{total}} = B_{\text{astro}} \times B_{\text{phot/SED}} \times B_{\text{temporal}} \times B_{\text{kinematic}}.$$

4. Computational infrastructure

HEALPix (Górski et al. 2005, *ApJ* 622, 759) tessellates the sphere into 12 base quadrilaterals :antCitation[]{citations="d86f567c-d353-495a-ac49-e41f198e1fb7"} recursively subdivided to $N_{\text{pix}} = 12\,N_{\text{side}}^2$ equal-area pixels of $\Omega_{\text{pix}} = 4\pi/N_{\text{pix}} = \pi/(3N_{\text{side}}^2)$ sr. The NESTED scheme — pixel p at order $k+1$ has children $\{4p, 4p+1, 4p+2, 4p+3\}$ — turns spatial range queries into integer comparisons; SQL B-tree indices on `ipix` become spatial indices for free. **NUNIQ** indexing $u = 4N_{\text{side}}^2 + p$ underpins the IVOA **MOC** (Multi-Order Coverage) standard now at v2.0 with time and spectral axes. :antCitation[]{citations="a7f19b3a-0839-496f-b866-73210ad74150,c4d9db3f-d02d-4cd7-b6bf-187c46ffa08b"} KD-trees and ball-trees on unit-vector projections give $O(N \log N)$ pair-match cost via the bijection $\theta = 2\arcsin(d_E/2)$.

The production stack: **CDS X-Match Service** (Boch, Pineau, Fernique 2012, 2020) at xmatch.cds.unistra.fr handles 10^9-row × 10^4-row matches in seconds via HEALPix-partitioned binary indices and per-pixel KD-trees; **TOPCAT/STILTS** (Taylor 2005, 2023) implements `tskymatch2`/`tmatch2`/`tmatchn` with multi-thread bin refinement; **NWAY** (Salvato 2018) in Python; **Astro Data Lab** at NOIRLab serves >2 PB of imaging (ScienceDirect) plus precomputed cross-matches like `ls_dr9.x1p5__tractor__gaia_dr3__gaia_source`. (Noirlab) IVOA standards **TAP/ADQL/Cone Search/MOC/HiPS** form the interoperability layer; **Lasair**, **ALeRCE**, **ANTARES**, **Fink**, **AMPEL**, **Babamul**, and **Pitt-Google** are the seven Rubin community brokers, (Oxford Academic) (Rubin Observatory) each performing real-time cross-match against Gaia/WISE/SDSS/variable-star catalogues at alert ingest. Recent infrastructure work: HLC2 (Zhang et al. 2023, heterogeneous CPU+GPU); (arxiv) **learned-index cross-match** (arXiv:2504.10931, 2025) replacing per-tile KD-trees with small neural networks for ~4× speedup; (arxiv) catsHTM (Soumagnac & Ofek 2018) for ms-scale local cone search. (arxiv)

5. Modern ML approaches

AstroCLIP (Parker, Lanusse, Cranmer et al. 2024, *MNRAS* 531, 4990; arXiv:2310.03024) trains separate transformer encoders (Oxford Academic) for Legacy Survey grz images and DESI spectra, (Semantic Scholar) aligned via InfoNCE loss

$$\mathcal{L} = -\frac{1}{N}\sum_i \log \frac{\exp(\text{sim}(z_i^I, z_i^S)/\tau)}{\sum_j \exp(\text{sim}(z_i^I, z_j^S)/\tau)}$$

with τ a learned temperature. Cosine similarity in the resulting embedding *is* a soft cross-matcher when astrometry is degraded. The **Multimodal Universe** collaboration (Lanusse, Parker, Smith, Cranmer, Walmsley et al. 2024, NeurIPS, arXiv:2412.02527) released 100 TB of curated multimodal data (OpenReview) on Hugging Face with explicit cross-match utilities to construct multimodal subsets (GitHub) — making "contrastive cross-match" tantalisingly close but not yet realised for point sources. **Stein et al.** (2022, *ApJ* 932, 107) (eScholarship) MoCoV2/SimSiam on 76 M Legacy galaxies; (eScholarship) **AstroDINO**, **PAPERCLIP** (Mishra-Sharma 2024, arXiv:2403.08851). (arXiv)

Graph-neural-network cross-match is emerging: Hong et al. (2026, Springer LNCS 16103) on VizieR catalogue recommendation with multi-relational GNN; **SIGNN** (Hepburn-Dickins et al. 2025, WACV) for star-tracker identification with angular-separation edges. ALeRCE's Cabrera-Vives et al. (2024, *A&A*) **transformer alert classifier** won the ELAsTiCC LSST simulation campaign. (Astro)

6. Bayesian hierarchical models and simulation-based inference

Extreme deconvolution (Bovy, Hogg, Roweis 2011, *Ann. Appl. Stat.* 5, 1657) generalises Gaussian-Mixture EM to per-datum heteroscedastic covariances and projection masks, (Semantic Scholar) with EM updates

$$q_{nk} \propto \alpha_k \mathcal{N}(y_n | R_n \mu_k, R_n V_k R_n^\top + S_n)$$

— foundational for inferring deconvolved colour/parallax/motion population distributions when association probabilities are themselves missing data. **CondXD** (Wang et al. 2025, *MNRAS* 541, 2815) replaces per-bin XD with neural conditional density heads at ~10× speedup. Oxford Academic

Nested samplers: **dynesty** (Speagle 2020, *MNRAS* 493, 3132), :antCitation[]{citations="fad84a94-6c1c-41d6-a762-0fa50419179f"} **MultiNest** (Feroz et al.), **PolyChord** (Handley et al. 2015, slice-sampling — best for >30 dim). Variational inference: **StarNet** (Liu et al. 2023, *JMLR* 24) :antCitation[]{citations="44723a6b-a4a4-4caa-8873-7589f922a619"} trains amortised variational catalogues for crowded fields. **BLISS** (Regier et al. 2023–2024) handles set-valued posteriors over variable N.

Simulation-based inference is the regime change. The Cranmer–Brehmer–Louppe (2020, *PNAS* 117, 30055) review :antCitation[]{citations="d6232e63-f167-4566-9f35-b90829cbdc97,4f2d3d84-69be-4402-8c59-3c22d8c94818"} distinguishes (i) **Neural Posterior Estimation** training $q_\phi(\theta|x)$ via forward KL; (ii) **Neural Likelihood Estimation** approximating $p(x|\theta)$; (iii) **Neural Ratio Estimation** training $d(x, \theta)$ to discriminate joint vs marginal. Toolkits: `sbi` (Tejero-Cantero et al. 2020, *JOSS* 5, 2505; :antCitation[]{citations="5a3bc50d-e515-4f60-bdcf-1d18e6202513"} "sbi reloaded" Boelts et al. 2025, :antCitation[]{citations="06863d30-1e1f-4380-8992-42a60e762989"} *JOSS* 10, 7754), :antCitation[]{citations="37a9e80a-315e-4a0d-84bc-6734613965a4"} `lampe` (Rozet et al. 2024), `swyft` (Miller et al. 2021 — truncated marginal NRE), `ltu-ili` (Ho et al. 2024). :antCitation[]{citations="ed37854d-f577-4a01-b862-f0836f634f4b"} Astronomy applications: **SimBIG** (Hahn et al. 2023, *PNAS* 120, e2218810120; Lemos et al. 2024, arXiv:2310.15256); **Mishra-Sharma & Cranmer (2022, *PRD* 105, 063017)** use NPE on the Galactic Centre γ-ray excess :antCitation[]{citations="bee7e6f2-40ab-4f22-928e-3ac98653314d,843d0c46-38ae-495e-93a6-acae17606597"} with point-source counts as latent — directly cross-match-relevant. Normalising flows for posteriors: **DINGO** (Dax et al. 2021, *PRL* 127, 241103) for gravitational waves.

Probabilistic cataloguing with reversible-jump MCMC over variable N: **PCAT** (Daylan, Portillo, Finkbeiner 2017, *ApJ* 839, 4); multiband PCAT (Feder et al. 2020, *AJ* 159, 163); :antCitation[]{citations="3419e920-6f2b-4275-825a-bac19f8d8693"} **PCAT-DE** (2021–2022) for diffuse-emission templates. :antCitation[]{citations="a44ee774-a6f5-4bde-8abb-09e399ed54b5"} **Blast** (Jones, McGill, Manning, Gagliano et al. 2024, arXiv:2410.17322) is a production transient-host pipeline integrating positional cross-match + ML host priors + sbi-accelerated SED fits — the closest existing template for a unified probabilistic framework.

7. Selection functions and correlated biases

The selection function $S(q) \equiv P(\text{source} \in C \mid q)$ enters the Poisson likelihood as $\lambda(q) = \nu(q) S(q)$. The **gaiaverse / GaiaUnlimited** programme operationalised this for Gaia: CoG II (Boubert & Everall 2020, *MNRAS* 497, 4246) for DR2, CoG V (Everall & Boubert 2022, *MNRAS* 509, 6205) for EDR3 sub-samples, **selectionfunctiontoolbox** (Boubert & Everall 2022, *MNRAS* 510, 4626) using needlets × GP across $(G, G-G_{RP})$, **Cantat-Gaudin et al. (2023, *A&A* 669, A55)** for the DR3 parent SF calibrated against DECaPS, and **Castro-Ginard et al. (2023, *A&A* 677, A37)** for sub-sample SFs with $S_{\text{tot}} = S_{\text{parent}} \cdot S_{\text{subset}}$. **Storey-Fisher, Hogg, Rix et al. (2024, *ApJ* 964, 69)** built the **Quaia** quasar catalogue — 6.65 M Gaia × unWISE QSO candidates — explicitly to deliver a clean *intersection* selection function for cosmology, addressing the correlated-bias problem head-on.

The fundamental difficulty: S_A and S_B are not independent because both depend on shared latents (extinction $A_V(\ell, b)$, crowding density $\rho_*(\ell, b)$, scan-law footprint, time-of-observation). The Bayesian cross-match likelihood factorises only after conditioning on these latents:

$$P(d_A, d_B | q, \boldsymbol{\eta}) = S_A(q|\boldsymbol{\eta})\, S_B(q|\boldsymbol{\eta})\, L_A(d_A|q)\, L_B(d_B|q), \quad \boldsymbol{\eta} = (A_V, \rho_*, \ldots)$$

— a hierarchical model rather than a separable product. No production cross-matcher fully implements this conditioning.

8. Confusion regimes and probabilistic deconvolution

Crowded fields (galactic bulge, M2, ω Cen, LMC/SMC core) break the one-detection-one-source assumption. **PCAT** (Daylan 2017; Portillo 2017, *AJ* 154, 132) is the canonical transdimensional sampler with birth/death/split/merge proposals, going ~1.5 mag deeper than DAOPHOT on M2. Multiband PCAT (Feder 2020) is 0.4 mag deeper still and 500× faster. **PCAT-DE** (Butler & Feder 2021; Feder 2022) jointly fits diffuse-emission templates. **StarNet** trades RJ-MCMC for a one-shot encoder via amortised VI; **BLISS** (Regier 2023–2024) extends with neural set-valued posteriors. The Gaia DR3 special-region astrometry release (Gaia Collaboration 2023f, *A&A* 680, A35) extends standard pipelines to 9 dense regions; DR4 will generalise this to all dense regions.

9. Recent benchmarks

PLAsTiCC (Kessler et al. 2019, *PASP* 131, 094501; Hložek et al. 2023, *ApJS* 267, 25) used a **flat-prior weighted multi-class log-loss**:

$$L = -\frac{1}{\sum_j w_j} \sum_j w_j \frac{1}{N_j} \sum_{i \in j} \ln \hat{p}_{ij}$$

with kilonovae upweighted. **ELAsTiCC** (2022–2024) streamed 139,106,486 simulated alerts on 4,328,315 objects to five Rubin brokers — a far more realistic test, with host-galaxy associations and 30 model classes including never-observed predicted events. **Rubin DP0.2** (300 deg² of simulated DESC DC2) is the current playground; **DP1** (released 2025, real ComCam data) is the first on-sky DIA test; **DP2** is planned for Jul–Sep 2026.

Probability metrics:

- **Brier score**: $\mathrm{BS} = \frac{1}{2N} \sum_i \sum_k (C_{ik} - p_{ik})^2$, with Murphy (1973) decomposition into Reliability−Resolution+Uncertainty.
- **Multi-class log-loss**: $\mathrm{LL} = -\frac{1}{N} \sum_i \sum_k y_{ik} \ln \hat{p}_{ik}$.
- **Expected Calibration Error**: $\mathrm{ECE} = \sum_m \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|$ — central diagnostic for deep-learning over-confidence.

There exists *no widely adopted cross-match benchmark*: NWAY's XMM-COSMOS reproduction and the Chandra Source Catalog v2.1.1 cross-matches (Rots 2025) are the de facto reference comparisons.

10. Surveys: quantitative reference

Survey	Sources	Astrometry	Bands	Epoch	Volume
Gaia DR3 (Jun 2022)	1.811 G total; 1.468 G full 5-par	σ(ϖ)≈0.03 mas at G=15, 0.5 mas at G=20	G/G_BP/G_RP, G ≲ 21	J2016.0; data 2014–2017	~50 TB
Gaia DR4 (Dec 2026)	~2.5 G + epoch series	√2× DR3	+ BP/RP epoch + RVS	J2017.5; 66 mo	~500 TB
catWISE2020	1.890 G	PM ~20 mas/yr at W1=15	W1, W2	2010–2018, 12 epochs	—
AllWISE / unWISE	0.747 G / +0.7 mag deeper	—	W1–W4	2010+	—
ZTF DR22 (Sep 2024)	937 G detections, 4.97 G LCs	—	g, r, i; m_5σ≈20.5	2018+	—
TESS	200 k 2-min targets + FFIs 30/10/200 s	T-band 600–1000 nm	single	2018+	—

Survey	Sources	Astrometry	Bands	Epoch	Volume
Pan-STARRS DR2	10.7 G ObjectThin / 2.26 G unique	floor 16 mas to Gaia	g/r/i/z/y, m_5σ stack 21.3–23.3	2010+	1.6 PB
SDSS DR19 (Jul 2025)	first MWM data + eRASS1 VAC	—	u/g/r/i/z + spectra	2025+	—
2MASS PSC	0.471 G	70–100 mas	J/H/K_s	~2000	—
Catalina CRTS	~0.5 G LCs	—	V	2004+	—
Vera Rubin / LSST	First sci alerts 24 Feb 2026	10 mas at r=21 → 1 mas at r=24	u/g/r/i/z/y, m_5σ single 23.9–25	2026+, 10 yr	~500 PB total, 20 TB/night, ~10^7 alerts/night
Euclid	Q1: 63.1 deg²; DR1 Oct 2026: ~2300 deg²	—	VIS + Y/J/H + slitless	2023+	—
Roman	NET Sep 2026 launch	—	0.48–2.30 μm, 8 filters	2026+	—

The "Rubin–Euclid–Roman triangle" emphasised in the 2020 Decadal Survey is the operational substrate for the next decade of point-source cross-matching.

PART 2 — IDENTIFIED GAPS

The literature reviewed above exposes eight gaps where genuine novelty is achievable. Each is technically defensible (i.e., not just engineering) and demonstrably underexplored as of April 2026.

Gap 1 — Joint association × classification × novelty in one principled posterior. Every operational pipeline (NWAY, ARCHES, Marrese, MUWCLASS, Blast) treats these as a sequential cascade: associate first, classify the matched object, then optionally flag anomalies. Information flows only forward, and uncertainty does not propagate backward. A source whose photometric class would have *disambiguated* a borderline positional match cannot influence the match decision. The Pineau (2017) factorisation $B = B_{\text{astro}} \times B_{\text{phot}}$ already hints at the unification, but no framework currently lets the *class posterior* feed back into the *partition posterior* in a self-consistent computation.

Gap 2 — Learned generative priors replacing analytical population models. Both NWAY and Pineau §11 use empirical histograms or Gaussian-mixture surrogates for $p(\boldsymbol{\theta} \mid \text{class})$. Conditional normalising flows (zuko, nflows, FFJORD), score-based diffusion models, and CondXD (Wang et al. 2025) trivially replace these with $q_\phi(\boldsymbol{\theta} \mid \text{class}, \boldsymbol{\eta})$ — yet no published cross-matcher uses learned generative priors over the joint colour–magnitude–parallax–PM space conditional on class and Galactic latents. CondXD is the closest, applied only to QSO contamination.

Gap 3 — Streaming, online, calibrated cross-match for alert brokers. LSST will produce ~10^7 alerts/night within a 60-second budget. Brokers currently perform classical positional cross-match against static catalogues (Gaia, WISE) and append context as alert annotations. *Posterior probabilities* for association — properly marginalised over Gaia DR3 PMs, WISE detection uncertainty, and host-galaxy ambiguity — are not computed in the streaming hot path. Amortised neural posterior estimation makes this tractable, but no broker has deployed it for cross-match per se.

Gap 4 — Causal disentanglement of selection from astrophysics. GaiaUnlimited (2023) gave the community S_{parent} and S_{subset}, but the Pearl-style do-calculus framing — "what is the population $p(\boldsymbol{\theta})$ I would observe if I *intervened* to remove crowding/extinction/scan-law effects" — has not been applied to cross-matched samples. Storey-Fisher et al. (2024, Quaia) is the most explicit acknowledgement of correlated multi-survey selection; full causal-graph treatment with structural equation models or instrumental-variable identification is a wide-open frontier.

Gap 5 — Probabilistic deconvolution at scale in confused fields. PCAT scales poorly beyond ~10^5 pixels and StarNet's amortised VI has been demonstrated only on M2-class fields. There is no production-ready *catalogue-level* probabilistic deblending integrated with cross-match for LSST/Roman bulge fields. The LSST DM stack does deterministic deblending; Rubin alerts inherit that decision irreversibly.

Gap 6 — Anomaly detection in cross-matched feature spaces. ALeRCE's deep-SVDD (Pérez-Carrasco et al. 2023) detects anomalies in light curves; AstroCLIP enables similarity search. But "objects whose multi-survey *cross-match feature vector* (positional residuals, colour residuals, PM residuals from class predictions) is jointly anomalous across surveys" is an underexploited regime. Tabula-rasa novelty in the joint cross-survey embedding could surface unusual binaries, hyper-velocity stars, lensed quasars with photometric anomalies, or cataclysmic-variable populations not covered by class priors.

Gap 7 — Calibrated uncertainty in deep-learning cross-matchers. Razim, Cabrera-Vives, ALeRCE classifiers, transformer alert classifiers all suffer the standard DL miscalibration disease. The PLAsTiCC/ELAsTiCC literature shows ECE is rarely reported and almost never optimised. A Bayesian deep-ensemble or normalising-flow posterior with explicit calibration objectives (focal loss, label smoothing, temperature scaling, conformal prediction) for *cross-match decisions* — not just classification — has not been published.

Gap 8 — Multi-modal foundation models that ingest catalogues, images, spectra, and light curves jointly. AstroCLIP aligns images and spectra. Multimodal Universe assembled the dataset. But a foundation model whose tokens span *catalogue rows* (with positions and uncertainties), *cutout images*, *spectra*, and *light curves*, trained to make *cross-match a downstream zero-shot task*, does not yet exist for point sources. PAPERCLIP gestures at this for proposal text + images.

PART 3 — ORIGINAL PROPOSAL: FLINT

3.1 Three candidate names

Three candidates with mnemonics:

1. **FLINT** — *Flow-Learned Identification and Novelty in Time-domain cross-matching.* The mnemonic doubles as "flint and tinder" — the framework strikes a spark of identification across catalogues. **Recommended primary name.**
2. **HALO-PE** — *Hierarchical Amortised Linkage with Outlier-aware Posterior Estimation.* More academic, signals the SBI heritage.
3. **CASCADE-N** — *Causally-Structured Cross-match with Amortised Selection-aware Density Estimation, Novelty-extended.* Best if the causal-graph extension dominates.

The remainder of this section develops **FLINT** in detail.

3.2 What FLINT borrows and what is new

Borrowed: the Budavári–Loredo (2015) partition formalism over Π; the Pineau (2017) χ-match acceptance geometry as a *coarse candidate generator*; the NWAY (Salvato 2018) factorisation $B = B_{\text{astro}} \cdot B_{\text{phot}}$; the GaiaUnlimited selection-function machinery $S(q)$; the sbi/ltu-ili amortised neural posterior estimation toolchain; the AstroCLIP multimodal contrastive embedding pattern.

Novel: (i) a *single posterior* over the partition, the class, the astrophysical parameters, and a continuous novelty score, in which the class likelihood feeds back into the partition probability; (ii) **conditional normalising-flow priors** $q_\phi(\boldsymbol{\theta} \mid c, \boldsymbol{\eta})$ over $(M, \text{colour}, \varpi, \boldsymbol{\mu})$ replacing analytical/empirical priors; (iii) a *learned-likelihood-free* mode where forward simulation of detections through per-survey selection functions is the only model commitment; (iv) an explicit **novelty channel** as an additional class $c = \varnothing$ with a flow trained on a complement distribution (Out-of-Distribution flow or negation-of-classes prior); (v) **streaming amortisation** so the network outputs $P(\text{partition}, c, \boldsymbol{\theta}, \text{novelty} \mid D)$ in a single forward pass at LSST ingest rates; (vi) a **causal-graph prior** $p(\Pi, c, \boldsymbol{\theta}, \boldsymbol{\eta})$ that conditions on shared survey latents $\boldsymbol{\eta} = (A_V, \rho_*, \text{scan-law}, t_{\text{obs}})$ to disentangle astrophysics from selection.

3.3 Mathematical formulation

Let $\mathcal{D} = \{(d_{ic}, \boldsymbol{x}_{ic}, \sigma_{ic}, t_c, c)\}$ be detections from surveys $c = 1..C$. Latents are: a partition Π of detections into objects o, with each object having class $k_o \in \{1, .., K, \varnothing\}$ ($\varnothing$ is the novelty class), astrophysical parameters $\boldsymbol{\theta}_o = (\boldsymbol{\omega}_o, \varpi_o, \boldsymbol{\mu}_o, M_o, \boldsymbol{\zeta}_o)$ (direction, parallax, proper motion, absolute magnitude, intrinsic colours/SED amplitudes/variability descriptors), and survey latents $\boldsymbol{\eta} = (A_V(\hat{\boldsymbol{\omega}}), \rho_*(\hat{\boldsymbol{\omega}}), s_c(\hat{\boldsymbol{\omega}}, t_c))$ shared across objects in a HEALPix tile.

The **FLINT joint posterior** is

$$\boxed{p(\Pi, \{k_o\}, \{\boldsymbol{\theta}_o\}, \boldsymbol{\eta} \mid \mathcal{D}) \;\propto\; p(\boldsymbol{\eta})\, p(\Pi \mid \rho_*) \prod_{o \in \Pi} \Big[\pi(k_o)\, q_\phi(\boldsymbol{\theta}_o \mid k_o, \boldsymbol{\eta}) \prod_{(i,c) \in S_o} L_c(d_{ic}, \boldsymbol{x}_{ic} \mid \boldsymbol{\theta}_o, \boldsymbol{\eta})\, S_c(\boldsymbol{\theta}_o \mid \boldsymbol{\eta}) \Big] \prod_c \prod_{(i,c) \notin \bigcup S_o} N_c($$

where:

- $p(\Pi \mid \rho_*)$ is a Poisson-process prior over partitions whose intensity scales with local source density.
- $\pi(k_o)$ is a class prior including the novelty class $\varnothing$ with $\pi(\varnothing) = \epsilon \sim 10^{-3}$ tunable.
- $q_\phi(\boldsymbol{\theta} \mid k, \boldsymbol{\eta})$ is a **conditional normalising flow** trained on labelled archival samples; for $k = \varnothing$ it is a flow trained as $q_\phi(\boldsymbol{\theta} \mid \varnothing, \boldsymbol{\eta}) \propto \mathrm{base}(\boldsymbol{\theta}) / \sum_k \pi(k) q_\phi(\boldsymbol{\theta} \mid k, \boldsymbol{\eta})$ (a "complement flow", trained via importance reweighting).
- L_c is the per-detection likelihood — at minimum the χ-match Mahalanobis form propagated to epoch t_c with PM and parallax,

$$L_c(\boldsymbol{x}_{ic} \mid \boldsymbol{\theta}_o) = \mathcal{N}\Big(\boldsymbol{x}_{ic} \;\Big|\; \boldsymbol{\omega}_o + \boldsymbol{\mu}_o(t_c - t_0) + \varpi_o \boldsymbol{\Pi}(t_c),\; \boldsymbol{V}_{ic}\Big) \cdot \prod_b \mathcal{N}(m_{ic,b} \mid M_{o,b} + 5\log_{10}(1/\varpi_o) - 5 + R_b A_V, \sigma_{ic,b})$$

- S_c is the GaiaUnlimited-style selection function for survey c.
- $N_c(d_{ic} \mid \boldsymbol{\eta})$ is the no-counterpart background density for unmatched detections, set self-consistently from local source counts.

Two equivalent computational paths:

(a) Explicit hierarchical Bayes path. Marginalise $\boldsymbol{\theta}_o$ analytically where possible (Gaussian astrometric block) and numerically (flow log-density for the rest):

$$\mathcal{L}_o(k) = \int q_\phi(\boldsymbol{\theta} | k, \boldsymbol{\eta}) \prod_{(i,c) \in S_o} L_c S_c \, d\boldsymbol{\theta} \;\approx\; \mathbb{E}_{\boldsymbol{\theta} \sim q_\phi(\cdot | k, \boldsymbol{\eta})} \Big[\prod_{(i,c) \in S_o} L_c S_c \Big]$$

estimated by importance sampling with $N_{\mathrm{IS}} = 64$ flow draws. The **FLINT Bayes factor** for object o is then

$$B_o^{\mathrm{FLINT}} = \frac{\sum_k \pi(k) \mathcal{L}_o(k)}{\prod_{(i,c) \in S_o} N_c(d_{ic} \mid \boldsymbol{\eta})}$$

with the novelty-normalised class posterior $P(k | o, \mathcal{D}) = \pi(k)\mathcal{L}_o(k) / \sum_{k'} \pi(k')\mathcal{L}_o(k')$. The partition is sampled by reversible-jump MCMC (PCAT-style birth/death/split/merge) or solved at MAP via integer linear programming when the candidate graph from χ-match pre-filtering has bounded treewidth.

(b) Fully amortised SBI path. Simulate $(\Pi, \{k_o\}, \{\boldsymbol{\theta}_o\}, \boldsymbol{\eta}) \sim p$, push through forward simulators of each survey including selection, train an inference network ψ_θ to output

$$\psi_\theta : \mathcal{D}_{\mathrm{tile}} \mapsto (\hat{\Pi}, \{\hat{k}_o\}, \{\hat{\boldsymbol{\theta}}_o\}, \mathrm{novelty}_o)$$

via a transformer-over-detections architecture trained with a composite loss

$$\mathcal{L}_{\text{FLINT}} = \underbrace{\mathcal{L}_{\text{part}}}_{\text{set-prediction Hungarian-matched cross-entropy}} + \lambda_1 \underbrace{\mathcal{L}_{\text{cls}}}_{\text{focal cross-entropy with calibration}} + \lambda_2 \underbrace{\mathcal{L}_{\text{flow}}}_{-\log q_\phi} + \lambda_3 \underbrace{\mathcal{L}_{\text{nov}}}_{\text{OOD energy score}} + \lambda_4 \text{ECE}$$

with the **novelty score** computed as

$$\text{nov}(o) = -\log \sum_k \pi(k) q_\phi(\hat{\boldsymbol{\theta}}_o \mid k, \hat{\boldsymbol{\eta}})$$

— literally the negative log-evidence under the union of known-class flows. Calibration is enforced by an explicit ECE penalty $\mathcal{L}_{\text{cal}} = \sum_m \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$ and a **conformal prediction wrapper** that converts the network's calibrated probabilities into $1-\alpha$ coverage sets at deployment.

Causal extension. The graph $\boldsymbol{\eta} \to S_c \to d_{ic}$ and $\boldsymbol{\eta} \to \boldsymbol{\theta}_o \to d_{ic}$ shares $\boldsymbol{\eta}$ as a confounder. We make the structural assumption that $\boldsymbol{\theta}_o$ is conditionally independent of S_c given $\boldsymbol{\eta}$ (the GaiaUnlimited assumption), letting the do-calculus identify the *intervened* population $p(\boldsymbol{\theta} \mid \text{do}(\text{survey})) = \int p(\boldsymbol{\theta} \mid \boldsymbol{\eta})\, p(\boldsymbol{\eta})\, d\boldsymbol{\eta}$. FLINT outputs both the *observational* posterior (used for matching) and the *interventional* posterior (used for population science) as separate heads.

3.4 Why this is genuinely novel

The combination is not in the literature: probabilistic cataloguing (PCAT) gives Π, NWAY/ARCHES give $B_{\text{astro}} \times B_{\text{phot}}$, sbi gives amortisation, AstroCLIP gives multimodal embeddings, GaiaUnlimited gives S — but no published framework binds these together in a single posterior with novelty as a built-in latent and selection as a causal confounder. The closest existing work is **Blast** (Jones et al. 2024), which is a transient-host pipeline using sbi for SED fitting downstream of a deterministic host association — FLINT's design subsumes it as a special case (Π trivial, $C = 2$).

The "turn-on-its-head" move is reframing cross-match not as an *assignment problem* (Hungarian/ILP/MCMC over discrete Π) but as a **transformer-over-detections sequence-to-set inference**, with the class posterior, the novelty score, and the calibrated probability emerging from the same forward pass.

3.5 Validation and evaluation plan

Datasets (in increasing scale):

1. *Toy bench:* A simulated 1 deg² field with 10^4 stars + 100 quasars + 10 transients + 1 contrived novelty class (e.g., a fake hyper-velocity white dwarf with anomalous $\boldsymbol{\mu}$ vs ϖ).
2. *Reproduction bench:* XMM-COSMOS Bayesian counterparts vs NWAY (Salvato 2018), reproducing $\geq 95\%$ reliability.
3. *Real-data bench A:* Gaia DR3 × AllWISE × ZTF DR22 × 2MASS × Pan-STARRS DR2 in a 10 deg² field including the bulge edge — measure performance at the highest crowding ρ_* achievable before DR4.
4. *Streaming bench:* ELAsTiCC-2 alerts ingested with FLINT cross-matching against a reference catalogue, measured throughput in alerts/sec on a single A100.
5. *Anomaly bench:* Inject 100 synthetic "unknown class" objects into bench 3; report novelty-channel AUC.

Baselines: astropy `match_to_catalog_sky` (positional), NWAY (Bayesian + photometric), TOPCAT `tmatchn` (positional, multi-table), the Marrese DR3 cross-match for Gaia external (where applicable), AstroCLIP cosine similarity (soft cross-match baseline).

Metrics: completeness/reliability curves, per-class precision/recall, multi-class log-loss, Brier score, **ECE as a first-class metric**, conformal-prediction coverage at $\alpha = 0.1$, novelty-channel ROC-AUC, throughput in association/sec.

3.6 Honest difficulty assessment

Component	Difficulty	Who does it
Toy benchmark + flow priors over Gaia DR3 colour-magnitude conditional on class	**Weekend hobby (months 1–3)**	Victor solo
Reproduction of NWAY on XMM-COSMOS	**Weekend hobby (month 3)**	Victor solo
Single-tile FLINT in hierarchical-Bayes path (mode (a)) on ~10^2 objects	**Serious nights/weekends (months 4–9)**	Victor with code review from a collaborator
Causal extension with explicit do-calculus heads	**Requires statistician/astrostat collaborator**	Victor + IAG-USP/COIN
Fully amortised transformer (mode (b)) at scale	**Requires institutional GPU + co-author**	CBPF (Bom) or LIneA
Streaming integration with Fink broker	**Requires Fink-Brazil collaboration**	CBPF + Ishida (CNRS)
Anomaly-channel calibration on real LSST/ELAsTiCC alerts	**Multi-author paper, 12+ months**	Brokers consortium

Realistic split: **mode (a) on benches 1–3 is a single-author paper** in *Astronomy and Computing* or *MNRAS*. **Mode (b) + streaming + anomaly is a 4–6 author flagship paper** requiring CBPF/IAG-USP co-authorship and a dedicated GPU allocation.

3.7 Path to publication

1. **First paper (months 9–14)**: "FLINT-α: Conditional normalising-flow priors for probabilistic multi-catalogue cross-matching", single-author (Victor + 1 advisor as second), *Astronomy and Computing* or *MNRAS Methods*. Benches 1–3.
2. **Second paper (months 18–24)**: "FLINT: Amortised joint association, classification, and novelty detection at LSST scale", multi-author with CBPF + Fink-Brazil. Benches 4–5.
3. **Third paper (months 24–36)**: "Selection-causal demographics from FLINT cross-matched Gaia DR4 × Roman", with IAG-USP + GaiaUnlimited collaborators.

Realistic Brazilian collaborators: **Clécio Bom (CBPF / LAB-IA)** as primary advisor candidate — he runs CATS within Fink and is precisely where the streaming + anomaly + cross-match agenda lives in Brazil; **Laerte Sodré, Lilianne Nakazono, Claudia Mendes de Oliveira (IAG-USP / S-PLUS)** for the photometric prior + Gaia × WISE benchmarks; **Emille Ishida (CNRS, Brazilian)** for Fink integration and the COIN cosmostatistics network. International: **Tamás Budavári (JHU)** would be a natural reviewer and potential co-author given his foundational role; **Mara Salvato (MPE)** for NWAY comparison; **François-Xavier Pineau (CDS)** for ARCHES integration.

Conferences: **ADASS XXXVI** (1–6 Nov 2026, Perth) for the methodology paper; **SAB XLIX** (25–30 Oct 2026, Natal) for the Brazilian community; **Rubin Community Workshop 2027** for the streaming version; **IVOA Interop** (June/Nov 2026) for standards-track integration.

PART 4 — PRACTICAL PATH FOR VICTOR

4.1 Phase 1 weekend starter: Gaia × AllWISE × TESS in 1 deg²

Target field: 1 deg² centred on the open cluster **NGC 2516** (RA = 119.50°, Dec = −60.83°, J2000) — Galactic latitude high enough to avoid extreme crowding, low enough to have rich Gaia+WISE+TESS coverage, and with a known PM signature (cluster $\boldsymbol{\mu} \approx (-4.7, +11.2)$ mas/yr) so you can verify your match recovers it.

Step-by-step (one weekend):

```python
# After the mamba env in 4.2 is set up:
from astroquery.gaia import Gaia
from astroquery.ipac.irsa import Irsa
from astroquery.mast import Catalogs
from astropy.coordinates import SkyCoord
import astropy.units as u

centre = SkyCoord(ra=119.50*u.deg, dec=-60.83*u.deg, frame='icrs')

# 1. Gaia DR3
gaia = Gaia.cone_search_async(centre, radius=0.6*u.deg).get_results()

# 2. AllWISE via IRSA
wise = Irsa.query_region(centre, catalog='allwise_p3as_psd', radius=0.6*u.deg)

# 3. TESS Input Catalog via MAST
tic = Catalogs.query_region(centre, radius=0.6*u.deg, catalog='TIC')

# 4. Propagate Gaia to WISE epoch (~2010.5)
dt = 2010.5 - 2016.0
ra_p  = gaia['ra']  + (gaia['pmra']/3.6e6)  * dt / np.cos(np.deg2rad(gaia['dec']))
dec_p = gaia['dec'] + (gaia['pmdec']/3.6e6) * dt

# 5. KD-tree match Gaia(propagated) ⇔ WISE at 1.5"
# 6. Compute Budavári-Szalay B for each candidate pair
# 7. Plot CMD G vs G_BP-G_RP coloured by P(H|D)
# 8. Identify high-PM outliers ⇒ likely brown-dwarf candidates
```

Deliverables for the weekend: (i) a reproducible notebook showing the proper-motion correction matters (compare matched fraction at 0" vs full PM-propagated); (ii) the cluster CMD with cluster sequence highlighted; (iii) a list of "interesting" sources flagged as $P(H|D) < 0.5$ at any pair — these are the residuals that a future FLINT-style novelty channel would flag.

4.2 Jupyter setup: from "Python only" to research-ready

One-time install:

```bash
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Minifo
bash Miniforge3-*.sh
mamba create -n xmatch python=3.11
mamba activate xmatch
mamba install -c conda-forge jupyterlab=4 ipykernel ipywidgets numpy=1.26 \
  scipy=1.13 pandas=2.2 matplotlib=3.9 astropy=6.1 astroquery=0.4.7 \
  healpy=1.17 dustmaps=1.0 scikit-learn=1.5 pyarrow duckdb \
  jax jaxlib pymc=5 arviz h5py tqdm
pip install dynesty==2.1.* emcee==3.1.* corner gala==1.9.* lightkurve==2.4.* \
  sbi==0.22.* zuko nflows numpyro pyro-ppl hpgeom cdshealpix pyvo astroml
```

Why each matters: **astropy/astroquery** for the I/O substrate; **healpy/cdshealpix/hpgeom** for HEALPix indexing (CDS HEALPix is rust-backed and pip-clean if cfitsio gives trouble); **dustmaps** for SFD/Bayestar reddening before colour matching; **dynesty/emcee** for nested sampling and MCMC over BS08 Bayes factors; **pymc 5 / numpyro** for hierarchical match models; **sbi / zuko / nflows** for FLINT-style learned priors and amortised posteriors — *this is your novelty toolchain*; **gala** for proper-motion + galactic-dynamics aware cross-match; **lightkurve** for TESS light curves; **pyvo** for TAP/ADQL queries to S-PLUS/Gaia/VizieR via the IVOA standard.

HEALPix gotcha: on Linux without `libcfitsio-dev`, `pip install healpy` will compile from source and fail. The conda-forge channel pulls `cfitsio` automatically; on macOS Apple Silicon use Miniforge ARM. Alternative: stick to `cdshealpix` + `hpgeom` for everything that doesn't need spherical-harmonic transforms.

Project skeleton:

```
xmatch-research/
├── pyproject.toml          # poetry/hatch with pinned versions
├── environment.yml         # mamba env export --no-builds
├── src/xmatch/             # importable package
|   ├── priors.py           # flow priors q_phi(theta|k,eta)
|   ├── likelihoods.py      # Mahalanobis + photometric L_c
|   ├── selection.py        # GaiaUnlimited wrappers
|   └── pipelines/
├── notebooks/              # 01_eda.ipynb, 02_baseline.ipynb, ...
├── data/{raw,interim,processed}/
├── tests/
└── papers/
```

Add `pre-commit` (black, ruff, isort), `nbstripout`, and either `dvc` or `git-lfs` for catalogue files. Use `papermill` to parameterise notebook runs.

4.3 Learning resources, ordered

1. **Coursera "Data-Driven Astronomy"** (University of Sydney, T. Murphy) — Week 2 *is* a cross-match exercise; audit free. If unavailable, GitHub mirrors exist.
2. **Astropy tutorials** at `learn.astropy.org` — Coordinates, Tables, "Cross-matching catalogs with astropy.coordinates".
3. **Rubin Data Academy** YouTube playlists (2024, 2025) — Rubin Science Platform, GCRCatalogs, photo-z.
4. **LSST DESC Tutorials** GitHub repo.
5. **CDS X-Match service notebook tutorials**.
6. **emcee / dynesty / sbi documentation tutorials** — work every example.

Reading list, in order (don't skip):

1. Sutherland & Saunders 1992, *MNRAS* 259, 413
2. Budavári & Szalay 2008, *ApJ* 679, 301 — read twice
3. Heinis, Budavári & Szalay 2009, *ApJ* 705, 739
4. Pineau et al. 2017, *A&A* 597, A89 — most useful starting point for FLINT
5. Marrese et al. 2017/2019/2022 — operational engineering reference
6. Salvato et al. 2018, *MNRAS* 473, 4937 — reproduce a result
7. Budavári & Loredo 2015, *Annu. Rev. Stat. Appl.* 2, 113 — the review
8. Wilson 2023, *MNRAS* 519, 5882 — PM-aware matching for LSST era
9. Cantat-Gaudin et al. 2023 + Castro-Ginard et al. 2023 (GaiaUnlimited)
10. Storey-Fisher et al. 2024 (Quaia) — correlated selection
11. Daylan, Portillo, Finkbeiner 2017, *ApJ* 839, 4 (PCAT)
12. Cranmer, Brehmer, Louppe 2020, *PNAS* 117, 30055 (SBI review)
13. Mishra-Sharma & Cranmer 2022, *PRD* 105, 063017 (NPE for unresolved sources)
14. Parker, Lanusse et al. 2024, *MNRAS* 531, 4990 (AstroCLIP)
15. Möller et al. 2021 + Förster et al. 2021 (Fink + ALeRCE — for the streaming context)

Books: Ivezić, Connolly, VanderPlas, Gray (2nd ed., 2020) *Statistics, Data Mining, and Machine Learning in Astronomy* is the single most useful text — chapters 4–6, 9, 10. Wall & Jenkins (2012) for hypothesis-testing intuition. Hilbe, de Souza & Ishida (2017) *Bayesian Models for Astrophysical Data* — partly authored by Emille Ishida (Fink). Gelman et al. *Bayesian Data Analysis* for foundations.

4.4 Brazilian collaborators and 18-month roadmap

Months 0–6 — competence building. Finish Coursera DDA + Astropy tutorials (M1–M2). Reproduce one figure of Salvato 2018 NWAY on XMM-COSMOS, push to public GitHub (M3).

Implement BS08 Bayes factor in JAX, benchmark vs NWAY (M4). Run S-PLUS DR4 × Gaia DR3 cross-match in one HEALPix tile (M5). Read Ivezić ch. 4–9. Write a 5-page memo identifying where current methods underperform — this defines your research question (M6).

Months 6–12 — original work. Choose one focused FLINT slice (recommended: conditional flow priors for multi-band photometric cross-match, the highest-novelty option leveraging Victor's Bayesian/forecasting background). Email *one* Brazilian collaborator with the GitHub repo as portfolio (M7). Implement on a single HEALPix order-5 tile (M8–9). Calibration analysis with ECE and conformal prediction (M10). Methods + results draft, open-source code with Zenodo DOI (M11). **First arXiv preprint by M12**.

Months 12–18 — submission and network. Incorporate co-author feedback (M13). Submit to *MNRAS* or *Astronomy and Computing* (M14). SAB XLIX abstract (deadline late Apr 2026; meeting 25–30 Oct 2026 in Natal-RN) (M14). ADASS XXXVI abstract (1–6 Nov 2026, Perth) (M15). Apply to RCW 2027 (M16). Address referee report (M17). Paper accepted; second-paper draft in progress (M18).

Brazilian collaborators in priority order:

1. **Clécio R. Bom** (CBPF / LAB-IA, Rio) — *the* primary candidate. Runs CATS classifier within Fink, integrates Gaia/WISE cross-match in alert pipelines. Lab email `labia@cbpf.br`; personal site `debom.cbpf.br`. Directly aligned with FLINT's streaming + novelty agenda.
2. **Lilianne Nakazono** / **Laerte Sodré Jr.** (IAG-USP) — S-PLUS × Gaia QSO classification, exactly the photometric prior territory. Sodré contact via `astro.iag.usp.br/~laerte/`.
3. **Emille E. O. Ishida** (CNRS / LPC-Clermont, Brazilian) — Fink co-PI, COIN founder. International channel.
4. **Ricardo Ogando** (ON / CTI; DES DM) — for tooling and infrastructure; site `ricardogando.github.io`.
5. **Pedro Bernardinelli** (IAG-USP) — TNO/asteroid linkage if FLINT extends to Solar System.
6. **Claudia Mendes de Oliveira** (IAG-USP) — S-PLUS PI. Listed in S-PLUS papers as `claudia.oliveira@iag.usp.br`.

International: **Tamás Budavári** (JHU), **Mara Salvato** (MPE), **François-Xavier Pineau** (CDS), **Dan Foreman-Mackey** (Flatiron) for the SBI-emcee bridge.

Conferences with confirmed dates 2026–2027:

Conference	Dates	Location
EAS 2026	29 Jun – 3 Jul 2026	Lausanne
RCW 2026	27–31 Jul 2026	SLAC, Menlo Park
SAB XLIX	25–30 Oct 2026	Natal-RN
ADASS XXXVI	1–6 Nov 2026	Perth
IVOA Interop	June 2026 (Strasbourg), Nov 2026 (TBA)	rotating

Practical first email template (Portuguese for Brazilian recipients):

> Olá Prof. [Nome], sou engenheiro de dados (BigQuery/Python/SQL, métodos Bayesianos) interessado em contribuir como pesquisador independente em cross-match probabilístico multi-catálogo. Reproduzi recentemente o pipeline NWAY (Salvato+ 2018) em [campo X] — código aberto em [github link] — e gostaria de discutir uma extensão usando [conditional normalising flows / proper motions / Fink filter]. Seria possível agendar uma conversa de 30 min nas próximas semanas? Currículo Lattes: [link]. Obrigado.

Send to Bom first, two-week intervals before contacting the next.

Final practical advice: open-source from day one (Brazilian astronomers vet collaborators by GitHub before email reply); get ORCID + Lattes + ADS library in month 1; join LSST community forum and LSST Discovery Alliance Slack; use SAB membership (≈R$250/yr) for reduced fees and the Brazilian astronomy mailing list; treat your data-engineering rigor as the comparative

advantage — Brazilian astro is rich in domain knowledge but light on professional MLOps and probabilistic ML, which is *exactly* where FLINT lives.

Conclusion

The cross-match field has reached a transition point. Twenty years of Bayesian rigor — Sutherland-Saunders LRs through Budavári-Szalay through Pineau χ-match — gave the community a clean factorisation $B = B_{\text{astro}} \cdot B_{\text{phot}} \cdot B_{\text{temporal}} \cdot B_{\text{kinematic}}$ implemented in CDS X-Match, NWAY, ARCHES, and Marrese's Gaia external pipelines. The next twenty years belong to learned, amortised, selection-aware posteriors that ingest catalogues *and* images *and* spectra *and* light curves jointly, treat Π as a transformer-output set, and surface novelty as a first-class latent. **FLINT** sketches one principled way to bind these ideas, and the gap analysis identifies seven other doors of comparable size still open. Victor enters at a privileged moment: the tools (sbi 2.x, zuko, GaiaUnlimited, Multimodal Universe), the data (Gaia DR4, ZTF DR22, Rubin DP1/DP2, Euclid Q1, Roman pre-launch), and the Brazilian institutional substrate (CBPF Fink-Brazil, IAG-USP S-PLUS, LIneA LSST IDAC) are all in place. The hard part now is not infrastructure — it is the disciplined translation of an engineer's calibration intuition into Bayesian astrostatistics. That translation is the publishable contribution.